

INTRODUCTORY PAPER



2020 MATHEMATICS

TIME ALLOWED: 35 MINUTES

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How many smiley faces are there?

☐

3

☐

7

☐

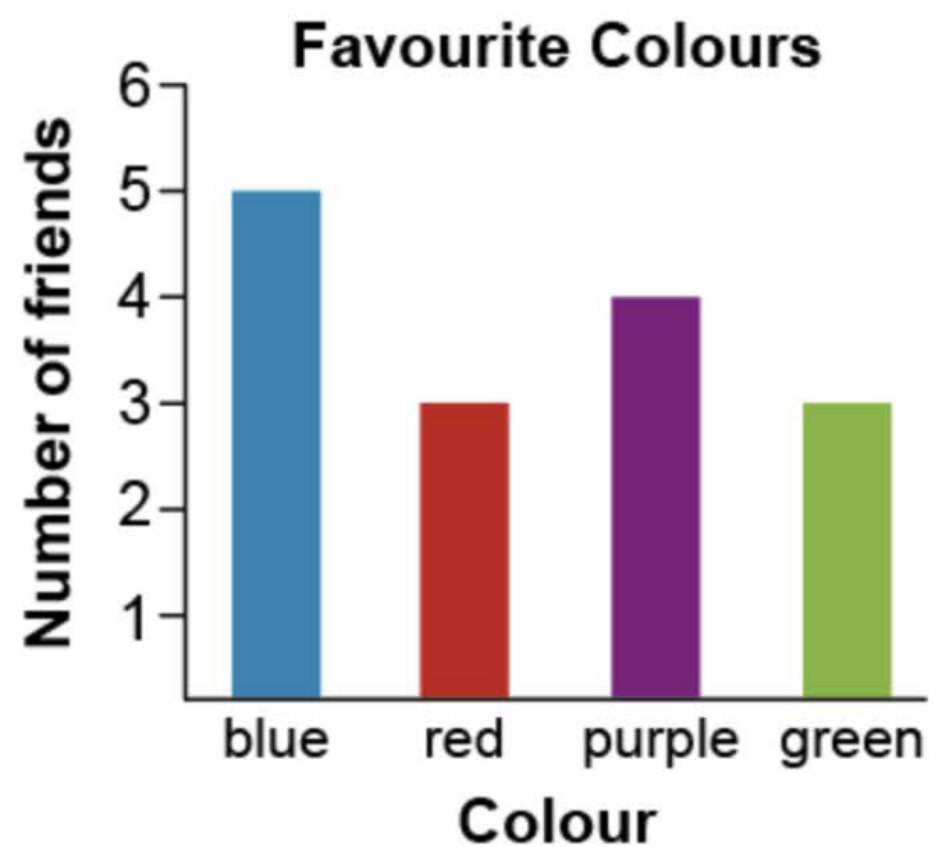
6

☐

8

**Back****ICAS.****Next**

Mary made this graph about her friends' favourite colours.



Which colour did Mary's friends like the most?

☐

blue

☐

red

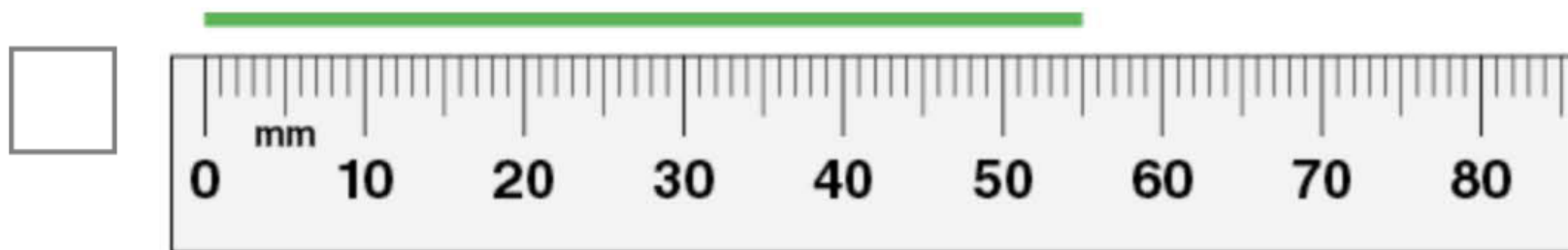
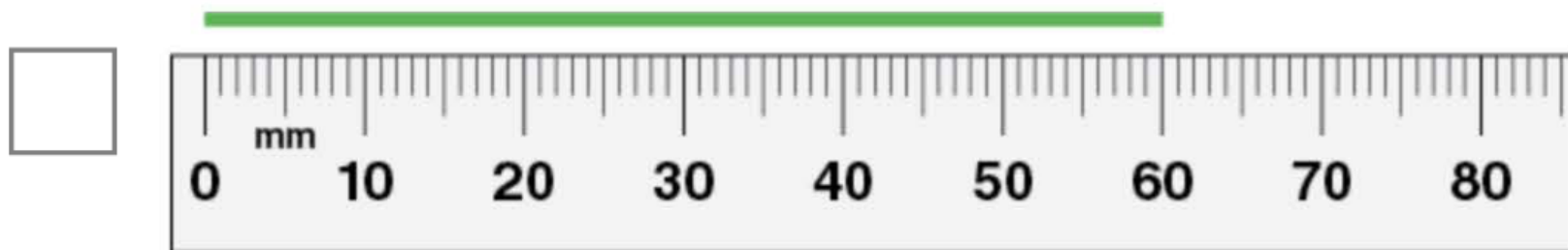
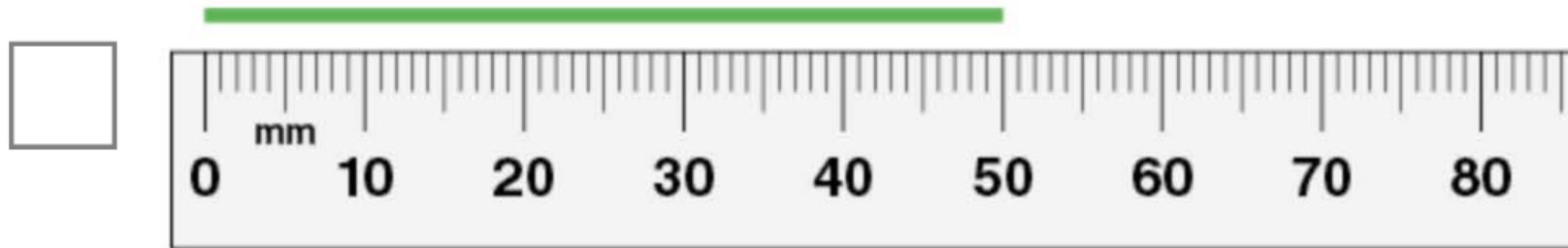
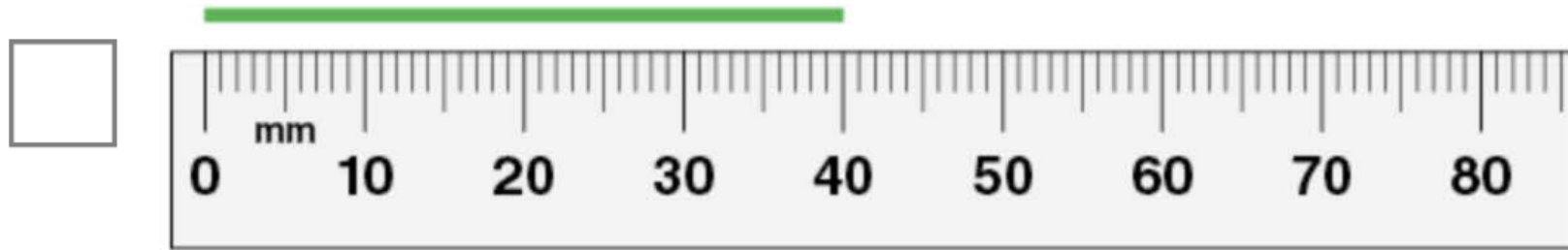
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purple

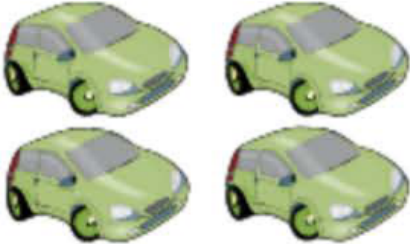
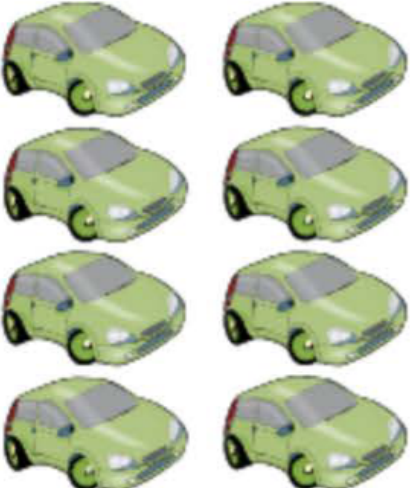
☐

green

Which line is 50 mm long?



Sunil made a pattern with his toy cars. The table below shows the steps in his pattern.

Step 1	Step 2	Step 3	Step 4
			

How many cars should there be in Step 3?

3

6

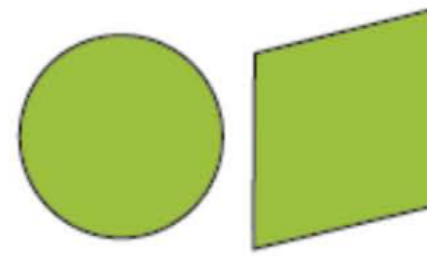
5

7

Emily put one shape on top of another shape. She drew around the edge to make this design.



Which two shapes did Emily have?

☐☐☐☐

Which one of these gives an answer closest to 18?

☐

$8 + 5$

☐

$8 + 9$

☐

$9 + 5$

☐

$9 + 7$



Simon made this pattern by counting up by the same number.

11, 16, 21, 26, 31, 

What is the next number in his pattern?

☐

42

☐

37

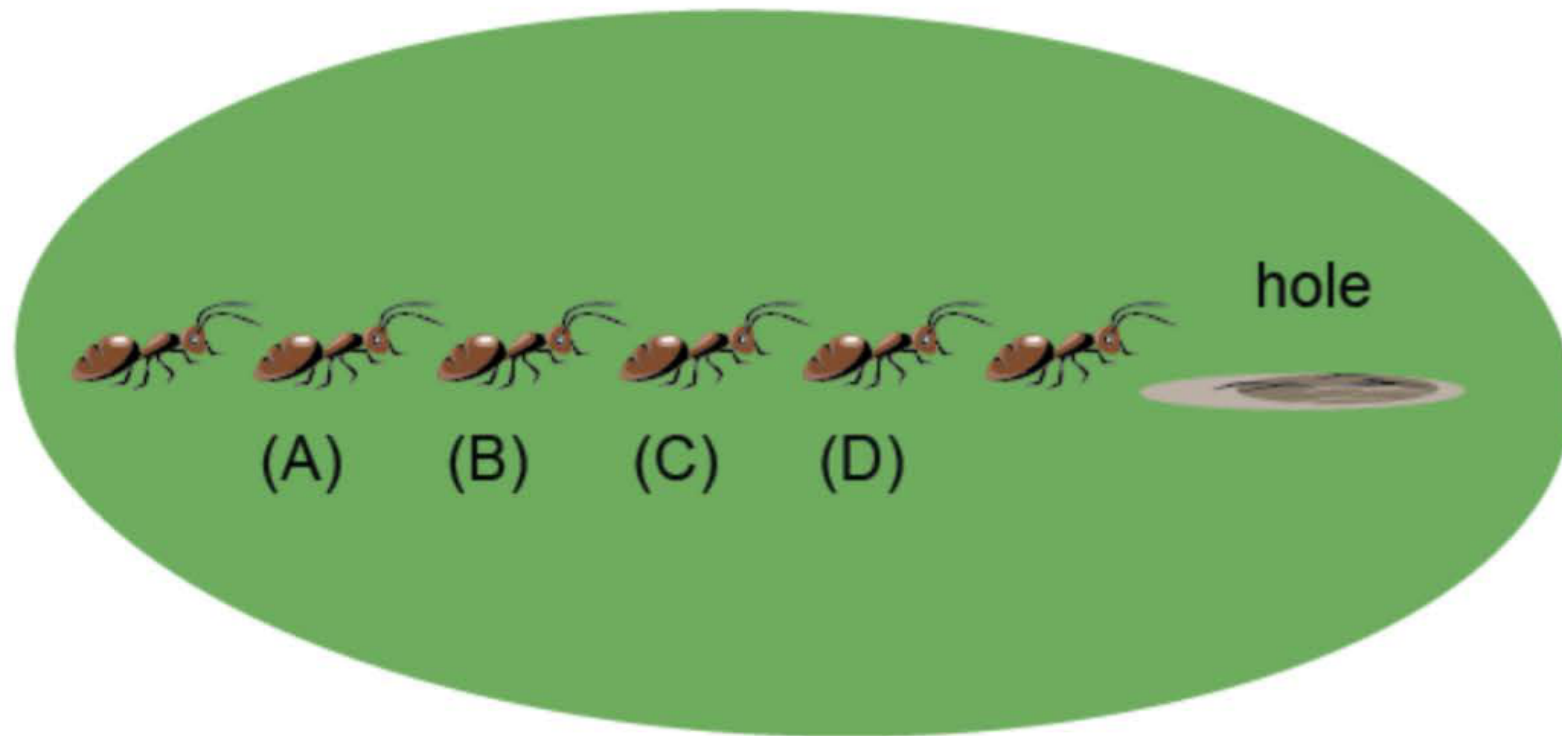
☐

36

☐

35

These ants are marching in a line.



Which will be the fifth ant to go into the hole in the ground?

☐

A

☐

B

☐

C

☐

D

Which of these was made by cutting a shape into equal parts?

☐☐☐☐

What number is halfway between 3 and 13?



7

8

9

10

Lori does a different after school activity for four afternoons every week. She has Art today.

Activity	Afternoon
Art	Tuesday
Dance	Friday
Music	Monday
Netball	Wednesday

What activity will Lori do tomorrow afternoon?

☐

Art

☐

Dance

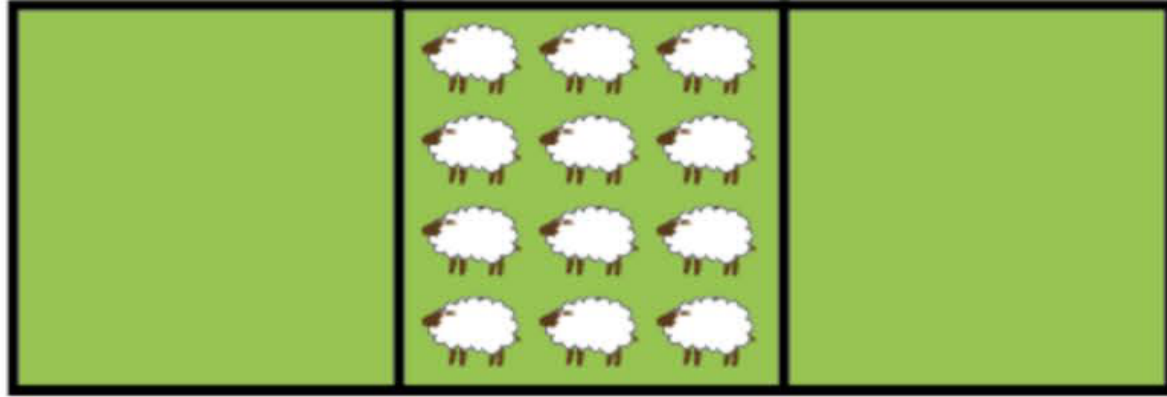
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Music

☐

Netball

Farmer Jim has three sheep paddocks.



He wants each paddock to have the same number of sheep in it.

How many sheep does he need to move in total?

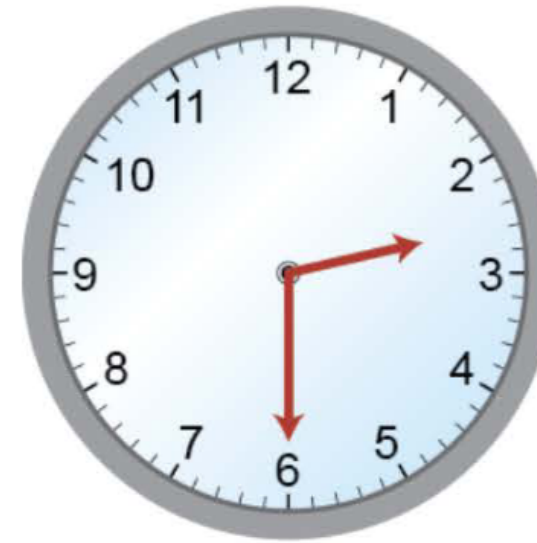
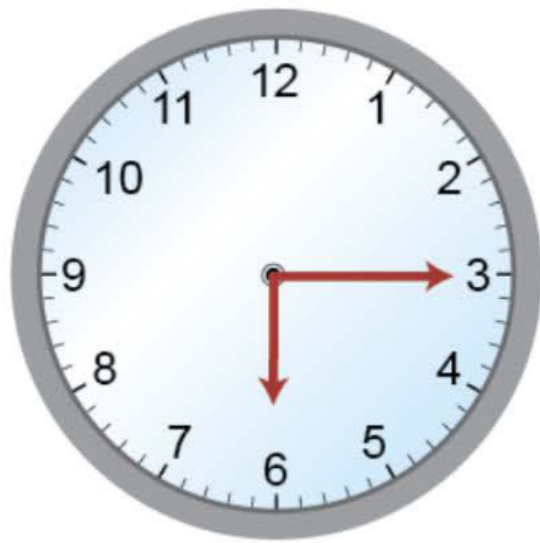
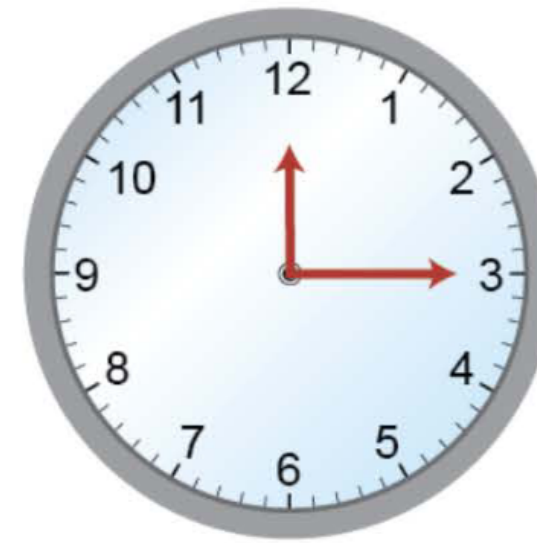
3

4

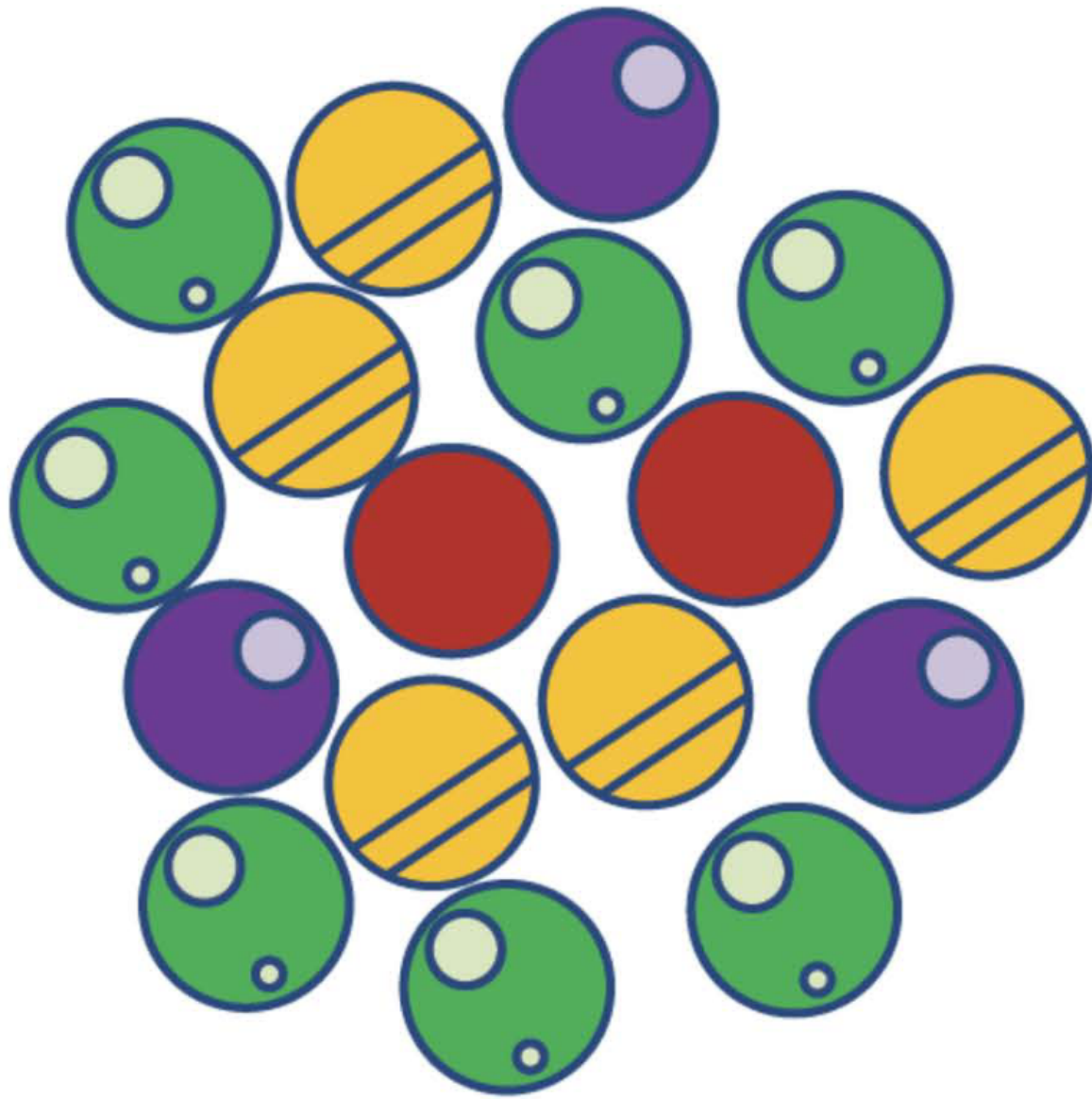
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8

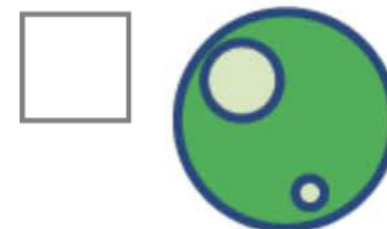
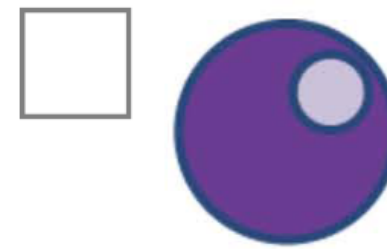
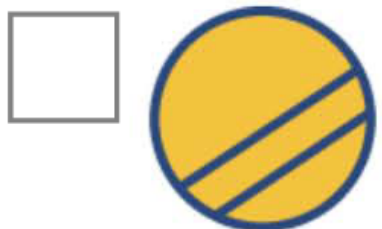
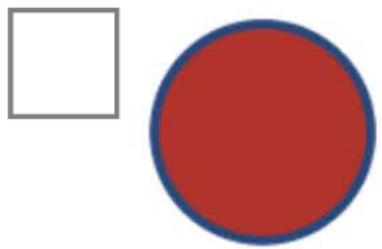
Which clock shows half past three?

☐☐☐☐

Ella picks one of these balls without looking.



Which colour ball is she **least** likely to pick?



Leon drew this number line.



What number must ? be?

☐

63

☐

70

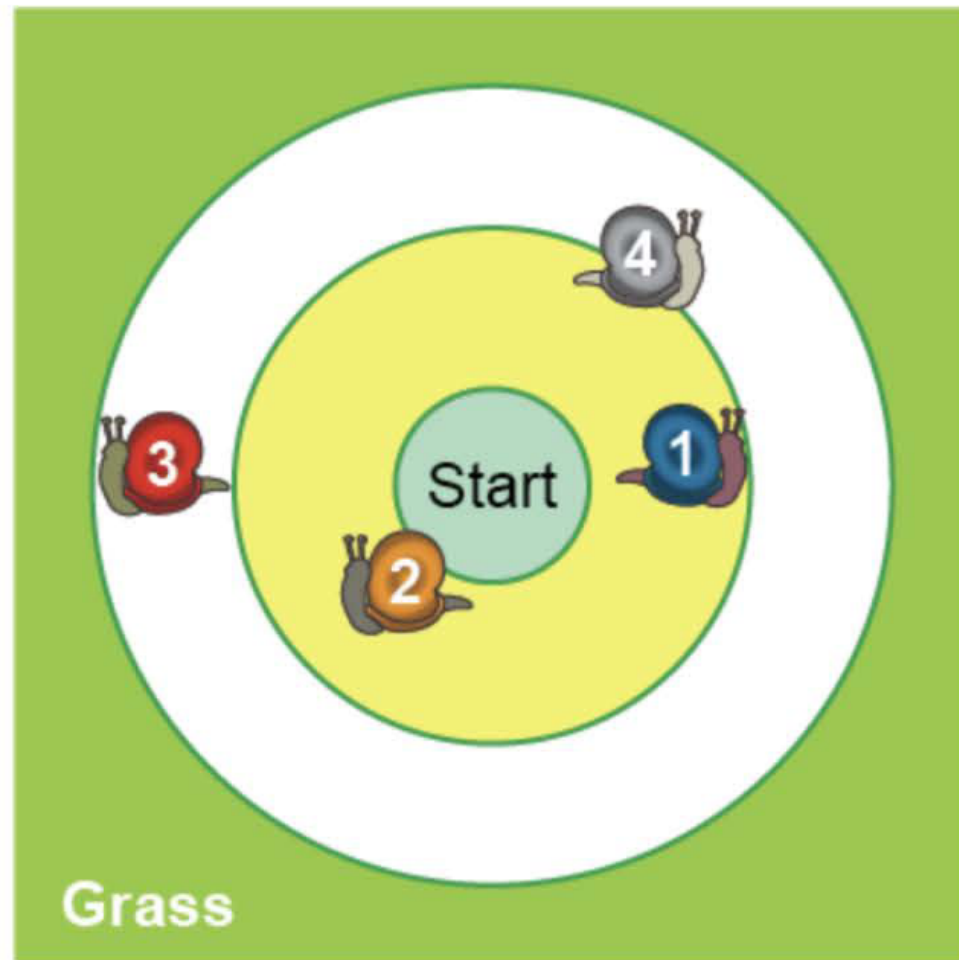
☐

75

☐

79

Four snails started in the centre circle. The picture shows their positions after 5 minutes.



Which list gives the order of the snails, from the closest to the grass to the furthest from the grass?

☐

3, 4, 1, 2

☐

3, 1, 4, 2

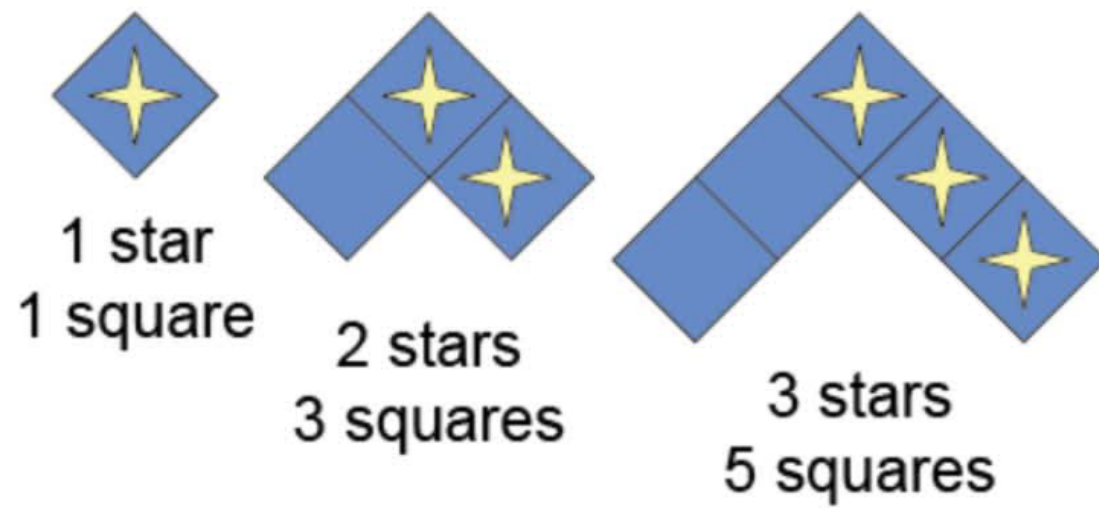
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1, 3, 4, 2

☐

1, 4, 3, 2

Hannah drew a pattern using shapes.



How many stars will a shape with 31 squares have?

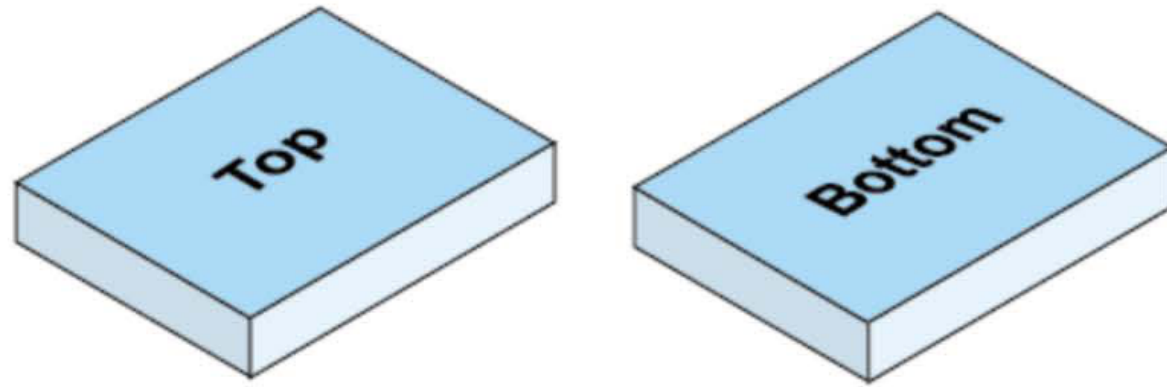
61

17

16

15

Here are two pictures of the same box.



How many faces does the box have?

☐

2

☐

3

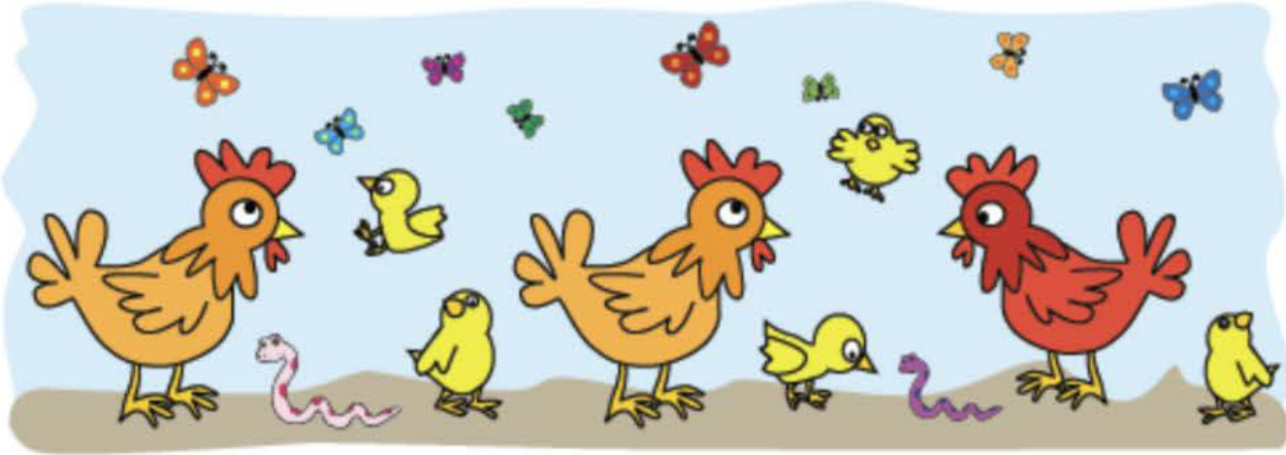
☐

6

☐

8

Rachel made a table to show how many of each type of animal there are in this picture.



She used one dot for each animal.

Which of these is Rachel's correct table?

☐

Animal	Number
butterflies	● ● ● ● ●
hens	● ●
chicks	● ● ●
worms	● ● ● ● ● ● ● ●

☐

Animal	Number
butterflies	● ● ● ● ●
worms	● ●
hens	● ● ●
chicks	● ● ● ● ● ● ● ●

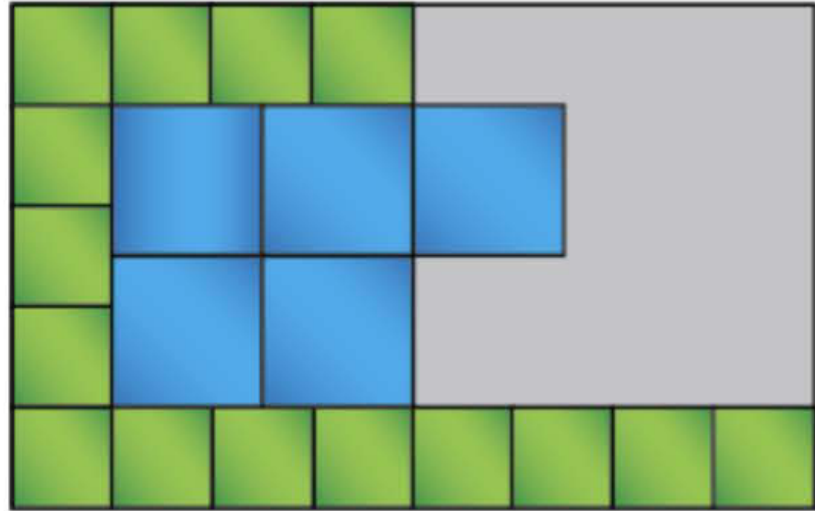
☐

Animal	Number
hens	● ● ● ● ●
worms	● ●
chicks	● ● ●
butterflies	● ● ● ● ● ● ● ●

☐

Animal	Number
chicks	● ● ● ● ●
worms	● ●
hens	● ● ●
butterflies	● ● ● ● ● ● ● ●

Gina started tiling a floor using large tiles and small tiles.
She used the small tiles around the outside and the large tiles in the middle, as shown.



How many tiles will Gina use altogether?

☐

28

☐

30

☐

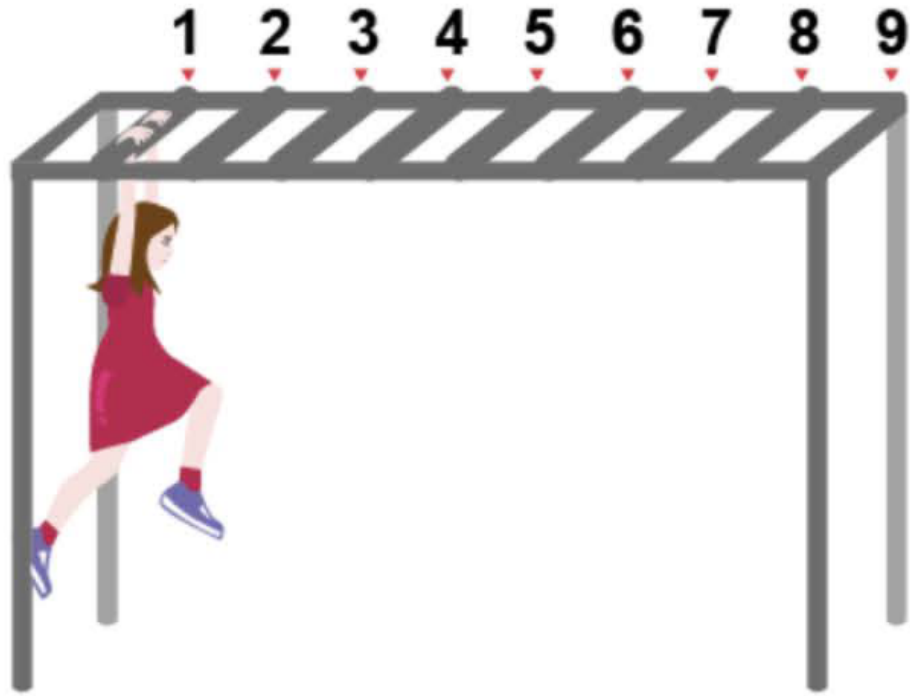
34

☐

40

Lily is swinging on the bars as shown. She starts with both hands on bar 1.

It takes her 2 seconds to move both hands, one at a time, to the next bar.



From which bar is she hanging after 8 seconds?

☐

4

☐

5

☐

7

☐

9

Here is a number pattern.

1, 2, 4, 7, **?**, 16

What number should **?** be?

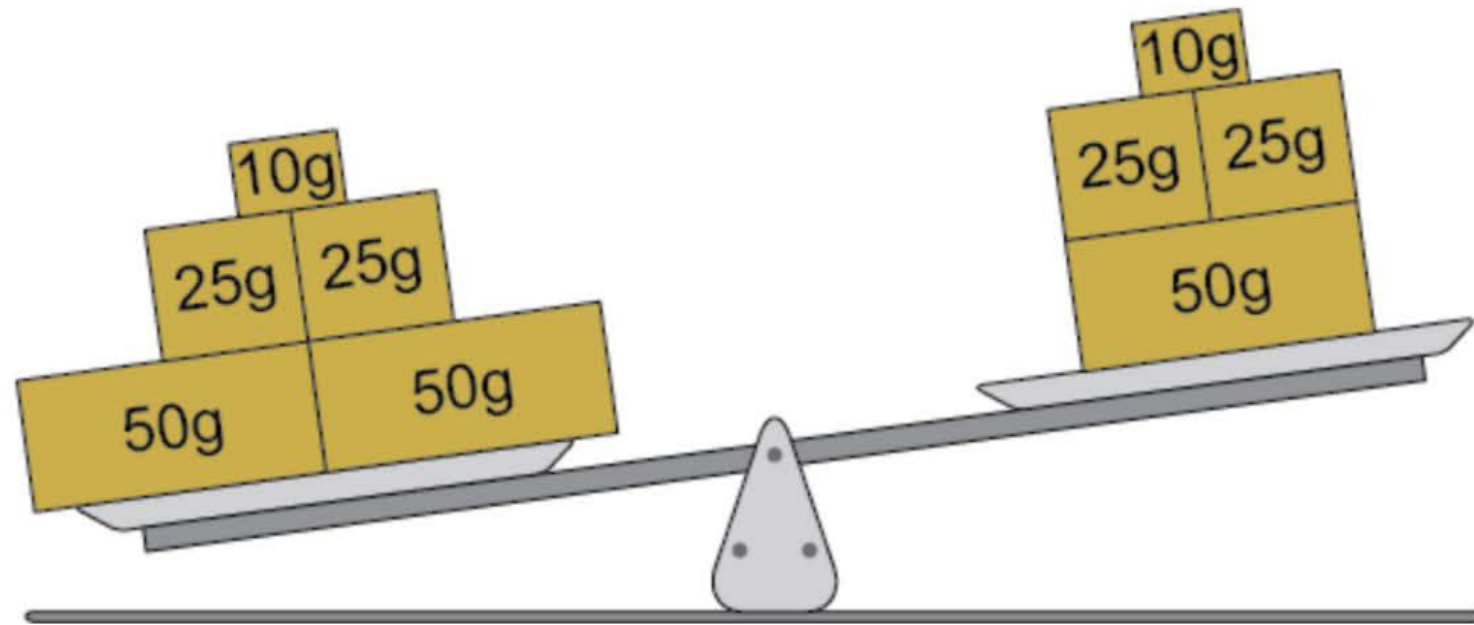
10

11

12

13

Joe placed some blocks on a scale.
He wanted the scale to be balanced.



Which weights must Joe add to the right-hand side to balance the scale?

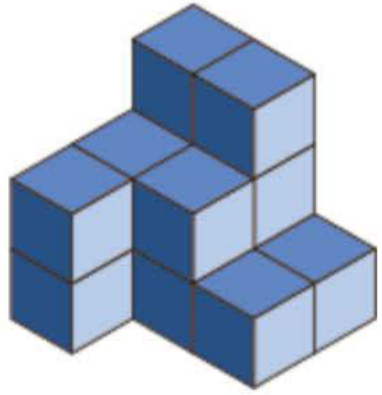
☐ $25\text{g} + 25\text{g}$

☐ $25\text{g} + 10\text{g}$

☐ $50\text{g} + 10\text{g}$

☐ $50\text{g} + 25\text{g}$

Tony stacked identical cubes to build this shape. He used 14 cubes.



How many cubes are in the bottom layer?

Half of Kerry's pencils are in her pencil case.

There are 10 pencils in the pencil case.

How many pencils does Kerry have?

☐

2

☐

5

☐

10

☐

20

*30 days has September,
April, June and November.
All the rest have 31,
Except for February which has 28 days,
or 29 in a leap year.*

What is the chance that any particular month has 31 days?

☐

certain

☐

likely

☐

unlikely

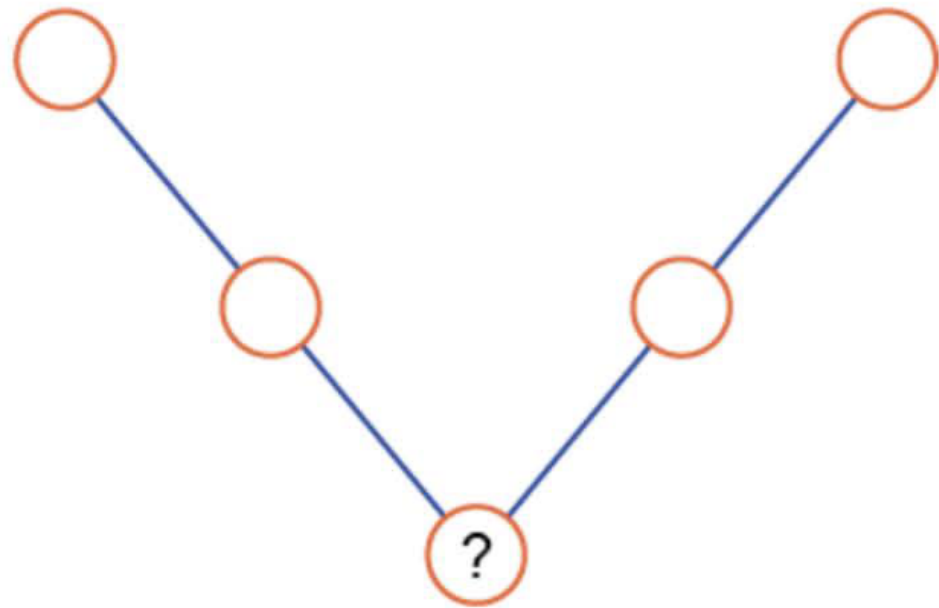
☐

impossible

Jim is completing a number puzzle using the numbers from 1 to 5.

Each number must be placed in a circle.

Each line of three numbers adds up to 8.



What number must **?** be?

☐

1

☐

2

☐

3

☐

4

☐

5

Amelia drew 9 dots on her page. Each dot is the same distance from the dots above, below, and to the side of it.



To draw a square Amelia must use a dot as a corner.

How many squares can she draw?

☐

4

☐

5

☐

6

☐

9

This is the rewards system at Liam's school:

1 Principal's Award	= 2 Gold Awards
1 Gold Award	= 2 Silver Awards
1 Silver Award	= 2 Bronze Awards
1 Bronze Award	= 5 Merit Awards

Liam has 30 Merit Awards so far.

How many more Merit Awards does he need to get **2** Principal's Awards? Type your number in the box.

Shelley made up a code.

Code	Value
	15
	40

She then wrote this line of code.



What is the value of Shelley's line of code?

☐

55

☐

66

☐

70

☐

150